

UNIT 1 COMPUTER SYSTEM

UNIT 1.2

Computer System and Buses



COMPUTER SYSTEM

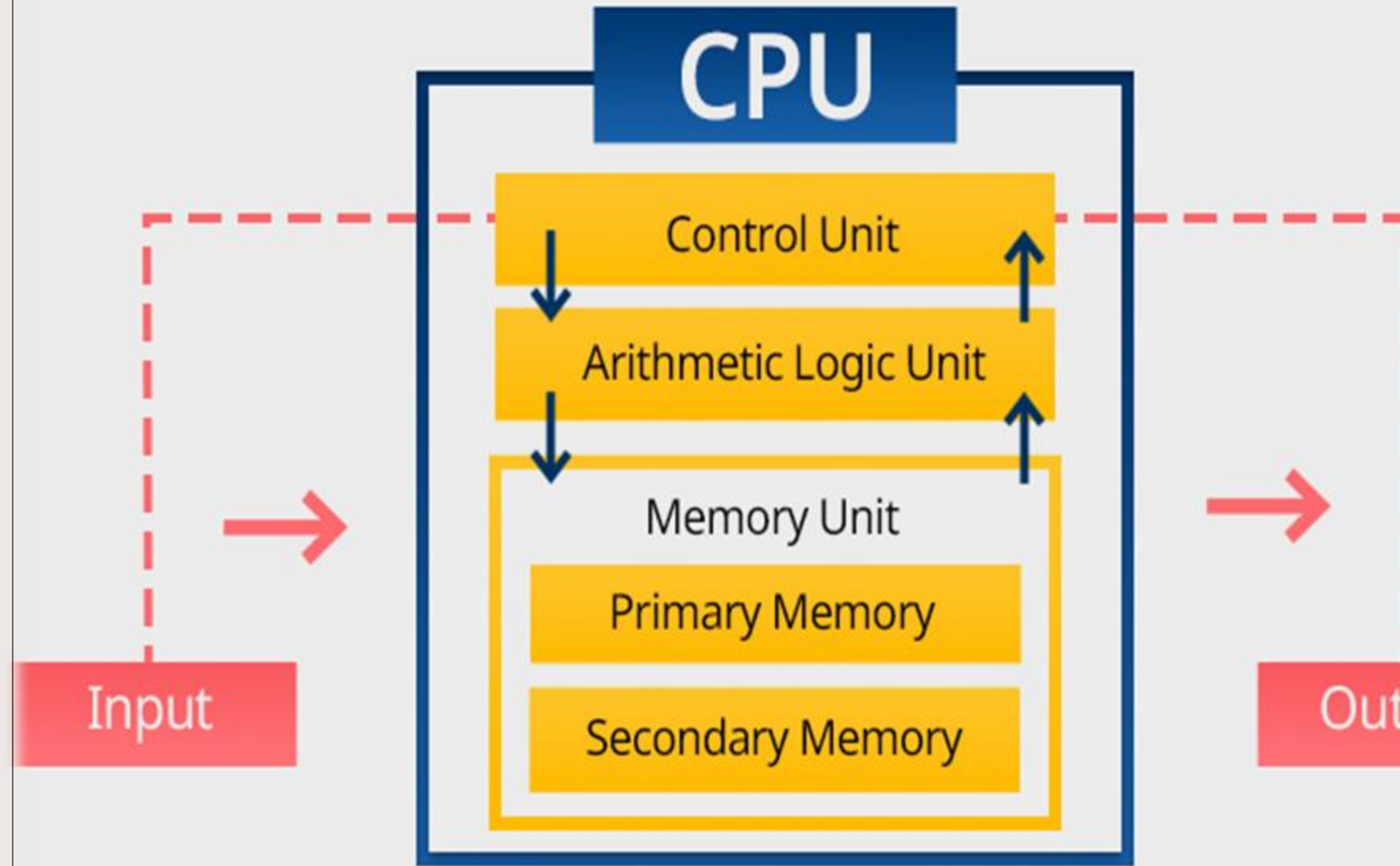


Fig: Computer System

INPUT UNIT

-> Definition:

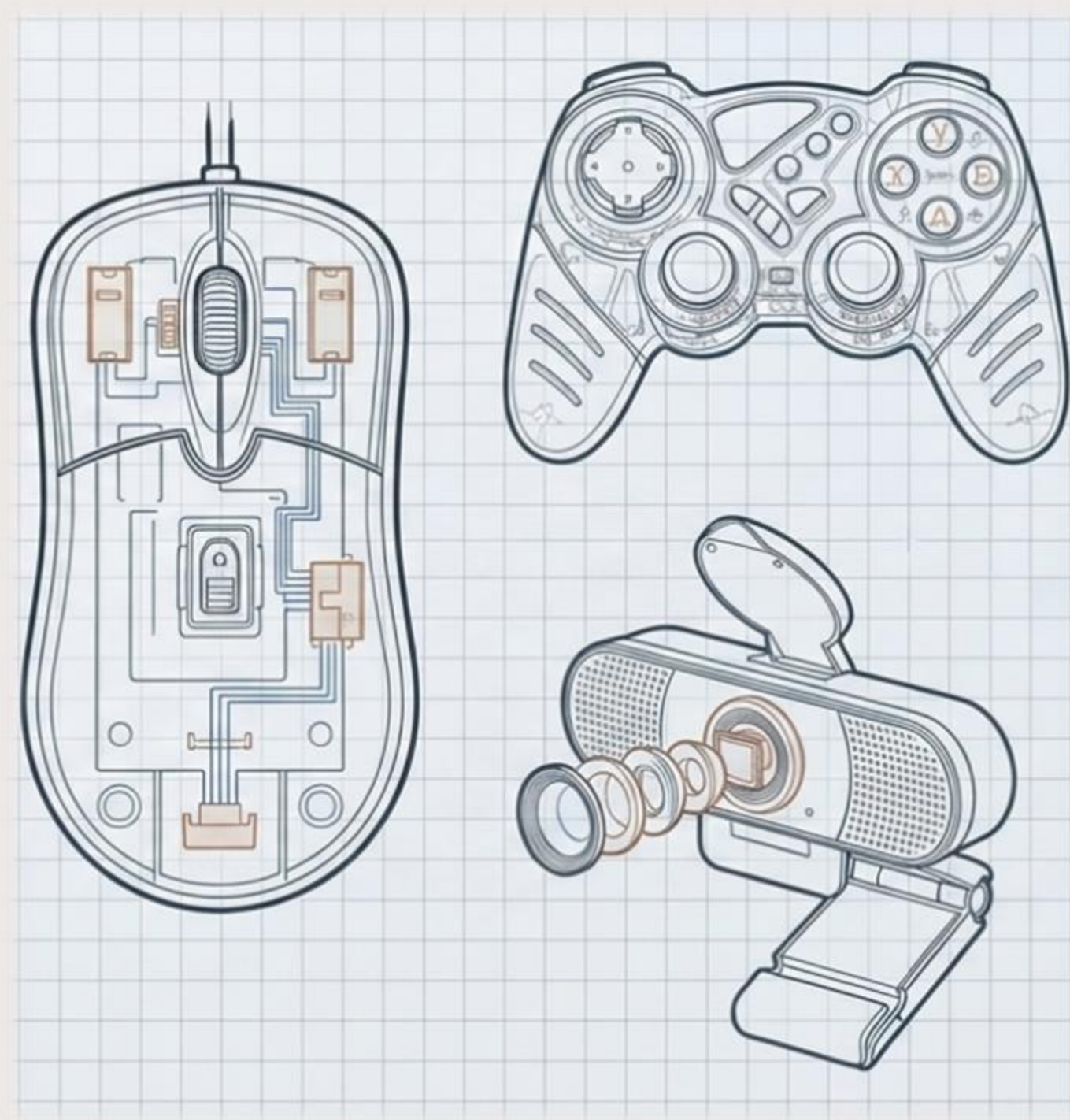
Those unit which gives data and instructions to the computer system is called input unit.

-> Functions / Operations:

1. It accepts or reads the instructions given by the user.
2. It converts received data into computer-understandable form.
3. It supplies converted data into primary memory.

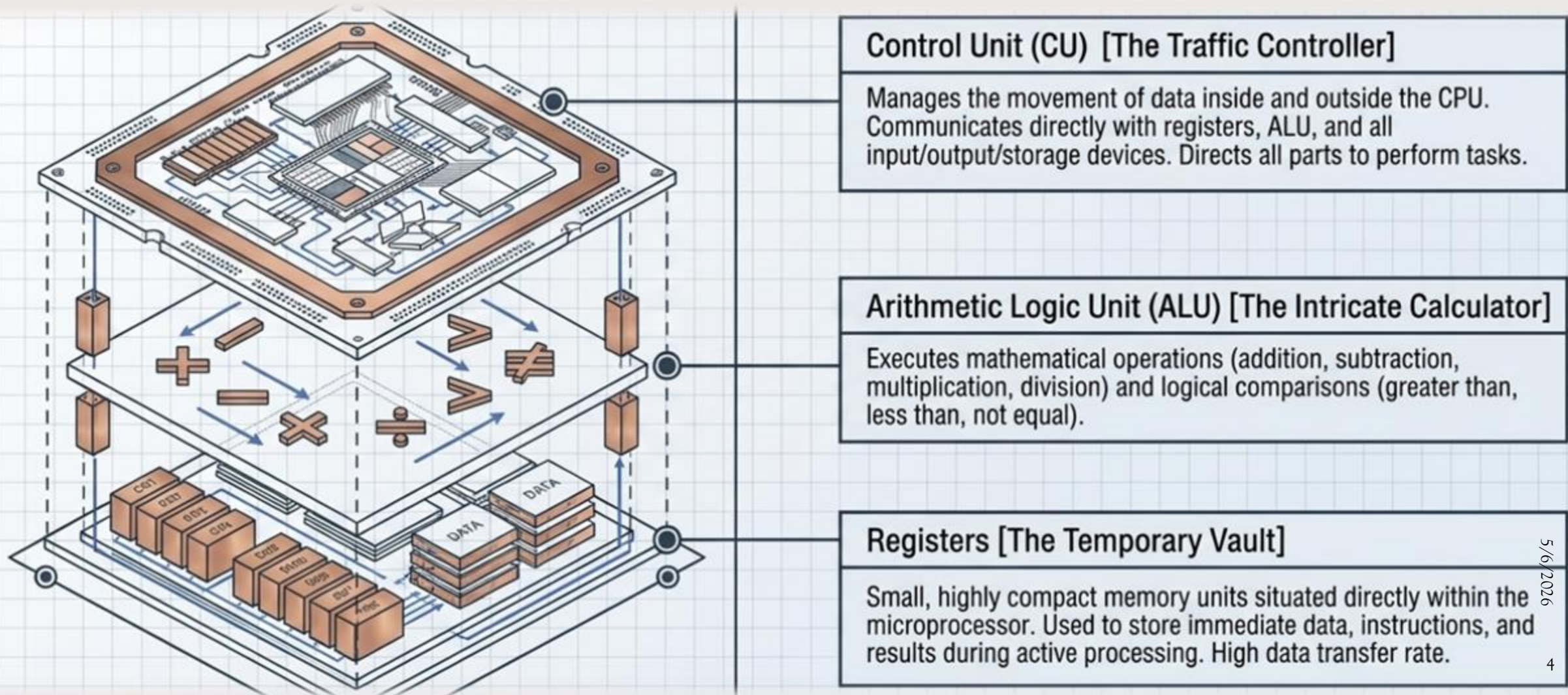
-> Examples:

1. Keyboard
2. Mouse
3. Scanner etc.



UNITS OF CPU

The 3 major units of CPU are shown in the figure:



CPU

-> Definition:

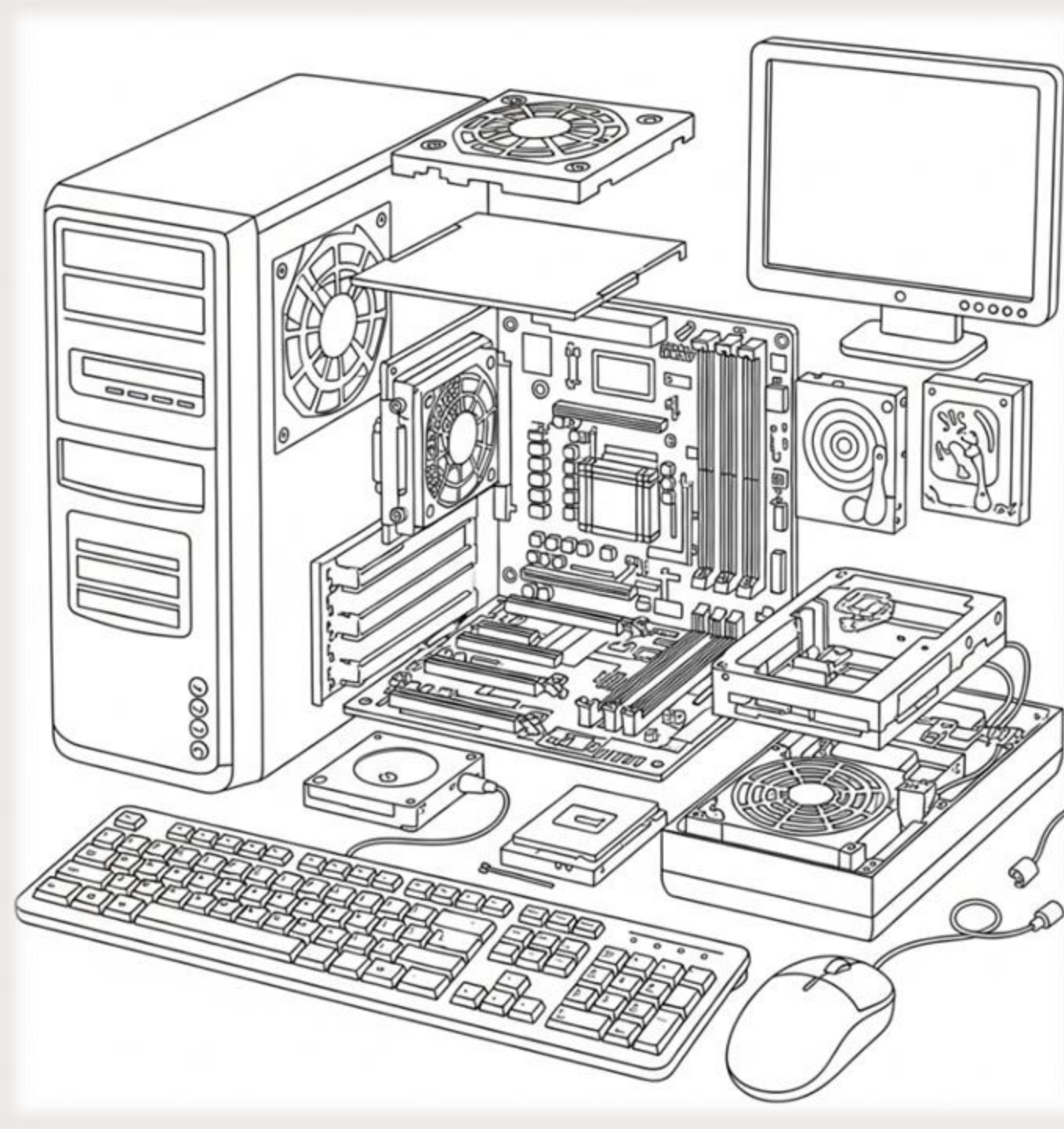
CPU stands for Central Processing Unit where all the processing of computer takes place.

-> Units of CPU:

1. Arithmetic Unit & Logical Unit
2. Control Unit
3. Memory Unit

-> Functions of CPU:

1. It Processes data at a very high speed.
2. It performs both arithmetic and logical operations.
3. It can hold data for the long period of time.



ARITHMETIC & LOGICAL UNIT (ALU)

-> Definition:

ALU stands for arithmetic and logical unit which performs all the operations of the computer system.

-> Function:

1. It performs arithmetic operations like addition, subtraction, multiplication & division.
2. It performs logical operations (AND , OR, NOT) and comparisons such as greater than, equal to, etc.

MEMORY UNIT

-> Definition:

Memory unit or primary memory is the part of CPU where data and instructions used by the computer is stored.

-> It is primary memory so it holds data for short period of time.

-> Functions:

1. It holds all the data that is required for processing.
2. It stores immediate result of processing.
3. It stores final result before it is sent to output device.

CONTROL UNIT

→ Definition:

A control unit is the unit of CPU which controls and co-ordinates the operation of entire computer system.

→ Function:

1. Sends input to input device and output to output device.
2. Decodes the instruction received from memory unit.

→ The speed of CPU is measured in Hertz (Hz). It is CPU clock rate or clock speed.

→ Nowadays, CPU speed is measured in Gigahertz (GHz).

1 GHz =	1000 Mega Hertz
	1,000,000 Kilo Hertz
	1,000,000,000 Hertz

The speed of CPU is 2.1 GHz

Now converting it into hertz,

$$\begin{aligned} 2.1 \text{ GHz} &= 2.1 * 1,000,000,000 \text{ Hz} \\ &= 2,100,000,000 \text{ Hz} \\ &\text{(2.1 Billion Hz)} \end{aligned}$$

OUTPUT UNIT

-> Definition:

Those unit which displays result generated by the computer system to the user is called output unit.

-> Types:

1. Hardcopy Output: Output that can be seen and heard. Eg:- monitor, speaker
2. Softcopy Output: Output that can be seen and touch. Eg:- printer

-> Functions / Operations:

1. It accepts the result produced by the computer i.e. in binary form.
2. It converts the binary form result in human understandable form.
3. It supplies converted result to the user.

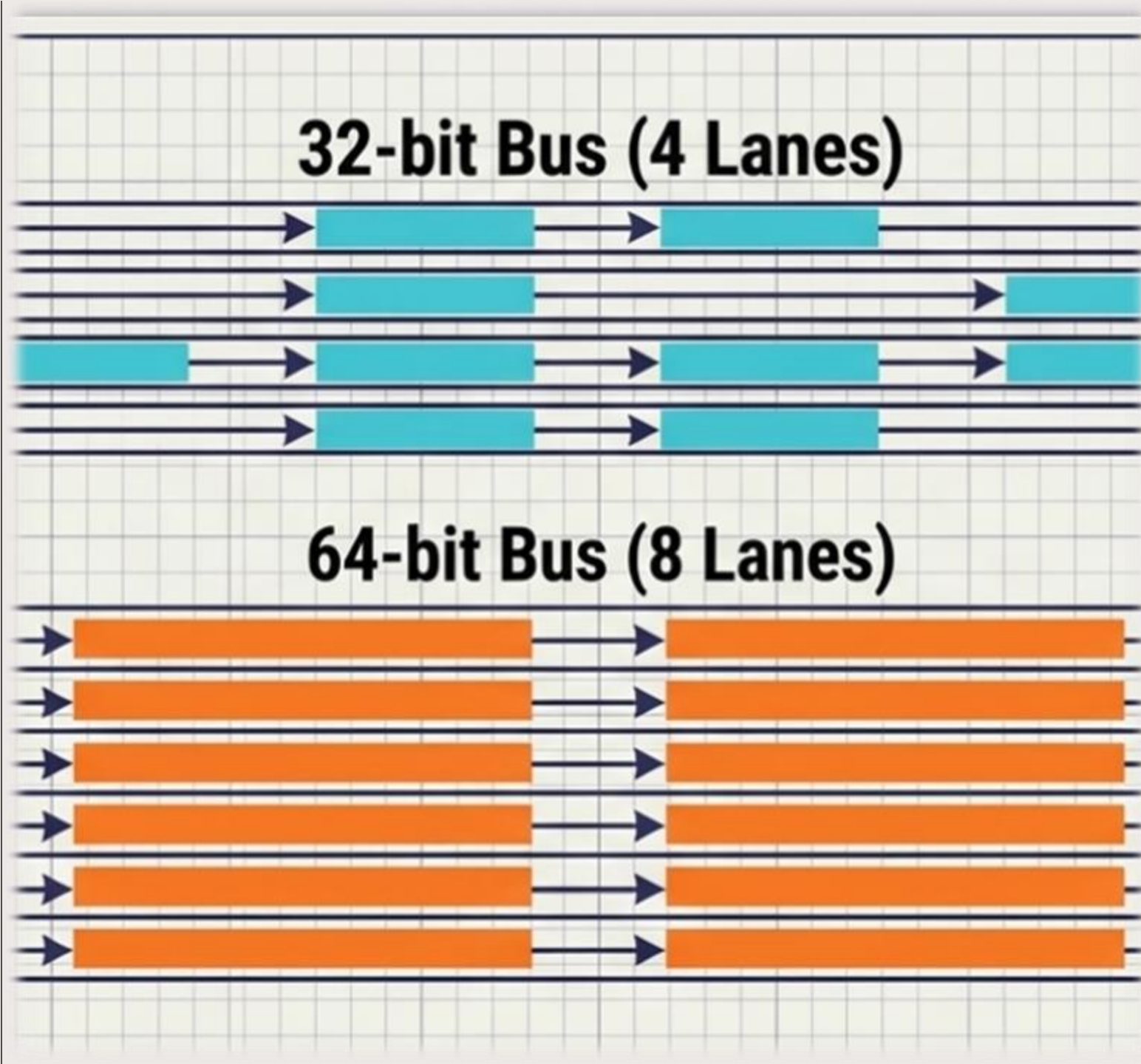


BUS

-> A computer bus is a communication system used to transfer data between different components of computer system.

-> Types of Bus:

- 1) Address Bus
- 2) Data Bus
- 3) Control Bus



TYPES OF BUS

Address Bus

- A collection of wires used to identify particular memory location in main memory is called Address Bus.
- It is unidirectional i.e. data flows in one directions only.

Data Bus

- A collection of wires or lines through which data is transmitted from one part of a computer to another is called Data Bus.
- These lines are bi-directional i.e. data flows in both directions.

Control Bus

- Carries control and timing signals between the CPU and other devices in the computer.
- It is a single wire.

IMPORTANT QUESTIONS

Short answer questions (2 marks)

1. What is input unit? Give 2 examples.
2. What is output unit? Give examples.
3. Define CPU with its units.
4. Define ALU. What are the functions of ALU.
5. What is control unit?
6. What is memory unit? Write about its types.
7. Define bus with its types.
8. Explain block diagram of computer system with figure.

Multiple Choice Questions (1 marks)

1. Which of the following is input device?
a) CPU b) Monitor **c) Keyboard** d) Printer
2. The speed of CPU is _____ hertz.
a) 2.1×10^9 Hz b) 2.1×10^8 Hz c) 2.1×10^7 Hz d) 2.1×10^{12} Hz
3. The example of output unit is _____.
a) Joystick **b) Monitor** c) Mouse d) Light pen
4. Which bus is used to transfer data from one component of computer to another?
a) Address Bus **b) Data Bus** c) Control Bus